

Chaehyeong Lee

Boulder, Colorado, USA

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Research Interests

Ocean dynamics and climate sciences: ocean heat budget; ocean's role in climate systems.

Geofluid dynamics: upper-ocean mixing processes in the frequency-wavenumber domain.

Education

University of Colorado Boulder

Ph.D. in Atmospheric and Oceanic Sciences

Advisors: Dr. Donata Giglio & Dr. Aneesh Subramanian

Boulder, CO, USA

Aug. 2024 – present

Yonsei University

M.S. in Atmospheric Sciences

Advisor: Dr. Hajoong Song

Seoul, Rep. of Korea

Mar. 2022 – Aug. 2023

Yonsei University

B.S. in Atmospheric Sciences

Seoul, Rep. of Korea

Mar. 2016 – Feb. 2022

Publications

In Progress.....

Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., & Molod, A. *Assessing the impact of sea surface salinity assimilation on extreme event prediction in NASA GEOS-S2S v2 model.*

Published.....

Lee, C., Song, H., Choi, Y., Cho, A., & Marshall, J. (2025). Observed multi-decadal increase in the surface ocean's thermal inertia. *Nature Climate Change*, **15**, 308-314. doi:10.1038/s41558-025-02245-w

Research Experience

Giglio's Research Group, CU Boulder

Research Assistant

Developing methods to improve NASA GEOS-S2S v2 simulations via sea-surface salinity assimilation; Filling gaps in upper-ocean mixing processes between observations and models through frequency-domain analysis.

Boulder, CO, USA

Aug. 2024 – present

Climate Processes and Predictability Group, CU Boulder

Research Assistant

Boulder, CO, USA

Aug. 2024 – present

Air-Sea Modeling Lab, Yonsei University

Research Assistant

Analyzed changes in the upper-ocean thermal state using SST observations; examined hysteresis of thermal memory under CESM 4×CO₂ experiments.

Seoul, Rep. of Korea

Dec. 2020 – Aug. 2024

Teaching Experience

Data Science Laboratory, taught by Prof. Donata Giglio at CU Boulder

Teaching Assistant

Boulder, CO, USA

Spring 2026

Climate & Civilization, taught by Prof. Yign Noh at Yonsei Univ.
Teaching Assistant

Seoul, Rep. of Korea
Spring 2023

Physical Oceanography, taught by Prof. Hajoon Song at Yonsei Univ.
Teaching Assistant

Seoul, Rep. of Korea
Fall 2022

Awards & Scholarships

2025: ATOC Fellowship, Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder
2025: Academic Research Grants (GCP research credits), Google LLC
2024: Outstanding Thesis Award, Yonsei University Graduate School, Yonsei University
2022–2023: Full tuition merit scholarship (for the top 2 graduate students), Yonsei University
2022: High Honors for academic performance, Yonsei University
2020–2021: Jillli Scholarship for academic performance, Yonsei University

Patent

Song, H., & Lee, C. (2025). *Evaluation system and method of persistence of SST anomalies using autocorrelation coefficient and arctangent regressive model*. Rep. of Korea Patent #1028135790000. doi:10.8080/1020220157159

Invited Talk

NASA Salinity Telecon

Virtual Meeting

Assessing the Impact of Satellite Sea Surface Salinity Assimilation on Vertical Structure of the Upper Ocean in the NASA GEOS-S2S 2.

Dec. 2025

Conferences

OSM 2026 (poster)

Glasgow, Scotland

Assessing the Impact of Satellite Sea Surface Salinity Assimilation on the Upper Ocean Thermal State in the NASA GEOS S2S-v2 Model.

Feb. 2026

Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., & Molod, A.

AGU Fall Meeting (poster)

Chicago, IL, USA

The increasing trend of persistence of sea surface temperature in the past 40 years.

Dec. 2022

Lee, C., Song, H., Cho, A., & Tak, Y.

Korean Society of Oceanography Spring Conference (talk)

Jeju, Rep. of Korea

Increasing persistence of SST anomalies and duration of marine heatwaves.

Jun. 2022

Lee, C., Song, H., Cho, A., & Tak, Y.

Workshops

Data Assimilation Training and Science Workshop

Norman, OK, USA

Consortium for Advanced Data Assimilation Research and Education & NOAA EPIC

June. 2026

User Training for the Glosea 6 Climate Prediction Model

Jeju, Rep. of Korea

Korea Meteorological Administration

Jan. 2022

Deep Learning Training: Fundamentals of Deep Learning

Gonju, Rep. of Korea

NVIDIA Deep Learning Institute

Jan. 2022

Service

Peer Reviewer: *Journal of Climate*

Technical Skills

Programming: Python (xarray, dask, Pangeo), Julia (Oceananigans)

HPC: Parallel/distributed computing, NCAR Casper/Derecho clusters

Tools: Git, Linux shell scripting, LaTeX